For the use only of a Registered Medical Practitioner

PRESCRIBING INFORMATION

SIMVOR TABLETS
(Simvastatin 10 / 20/ 40 mg)

COMPOSITION

SIMVOR TABLETS 10 mg
Each film-coated tablet contains
Simvastatin USP…………10mg
Inactive components: Microcrystalline Cellulose, Pregelatinised Starch, Magnesium Stearate, Purified Water, Butylated Hydroxyanisole, Isopropyl Alcohol, Low Subtituted Hydroxypropyl Cellulose, Opadry Pink OY-LS-54901, Lactose DC Grade and Lactose Monohydrate.

SIMVOR TABLETS 20 mg
Each film-coated tablet contains
Simvastatin USP…………20mg
Inactive components: Microcrystalline Cellulose, Pregelatinised Starch, Magnesium Stearate, Purified Water, Butylated Hydroxyanisole, Isopropyl Alcohol, Low Subtituted Hydroxypropyl Cellulose, Opadry Brown OY-LS-56504, Lactose DC Grade and Lactose Monohydrate

SIMVOR TABLETS 40 mg
Each film-coated tablet contains
Simvastatin USP …………40mg
Inactive components: Microcrystalline Cellulose, Pregelatinised Starch, Magnesium Stearate, Purified Water, Butylated Hydroxyanisole, Isopropyl Alcohol, Low Subtituted Hydroxypropyl Cellulose, Opadry Pink OY-LS-54902, PEG 400, Lactose DC Grade and Lactose Monohydrate

PRODUCT DESCRIPTION

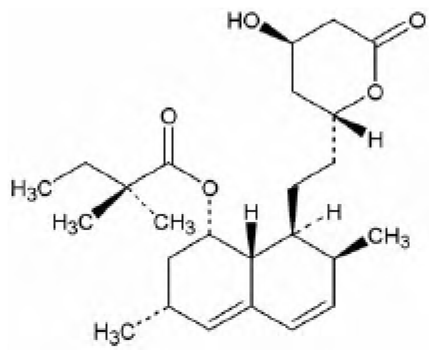
SIMVOR TABLETS 10 mg
Light pink coloured film coated oval shaped tablets, debossed with ‘10’ on one side and score line on the other side.

SIMVOR TABLETS 20 mg
Peach coloured film coated oval shaped tablets, debossed with ‘20’ on one side and score line on the other side.

SIMVOR TABLETS 40 mg
Pink coloured film coated oval shaped tablets, debossed with ‘40’ on one side and score line on the other side.

DESCRIPTION

Simvastatin is a lipid-lowering agent that is derived synthetically from a fermentation product of *Aspergillus terreus*. After oral ingestion, simvastatin, which is an inactive lactone, is hydrolyzed to the corresponding β-hydroxyacid form. Simvastatin is butanoic acid, 2, 2-dimethyl-,1,2,3,7,8,8a-hexahydro-3,7-dimethyl-8-[2-(tetrahydro-4hydroxy-6-oxo-2*H*-pyran-2-yl)-ethyl]-1-naphthalenyl ester, [1*S*-[1α,3α,7β,8β(2*S**,4*S**),-8aβ]].The empirical formula of simvastatin is $C_{25}H_{38}O_5$ and its molecular weight is 418.57. Its structural formula is:



SIMVASTATIN

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

- Pharmacodynamics

Simvastatin is a prodrug and is hydrolyzed to its active β-hydroxyacid form, simvastatin acid, after oral ingestion. Simvastatin is a specific inhibitor of 3-hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase, the enzyme that catalyzes the conversion of HMG-CoA to mevalonate, an early and rate limiting step in the biosynthetic pathway for cholesterol. In addition, simvastatin reduces VLDL and TG and increases HDL-C.

Elevated levels of total-C, LDL-C, as well as decreased levels of HDL-C are associated with the development of atherosclerosis and increased cardiovascular risk. Lowering LDL-C decreases this risk. However, the independent effect of raising HDL-C or lowering TG on the risk of coronary and cardiovascular morbidity and mortality has not been determined.

- Pharmacokinetics

Simvastatin is a lactone that is readily hydrolyzed *in vivo* to the corresponding β-hydroxyacid, a potent inhibitor of HMG-CoA reductase. Inhibition of HMG-CoA reductase is the basis for an assay in pharmacokinetic studies of the β-hydroxyacid metabolites (active inhibitors) and, following base hydrolysis, active plus latent inhibitors (total inhibitors) in plasma following administration of simvastatin.

Following an oral dose of ^{14}C-labeled simvastatin in man, 13% of the dose was excreted in urine and 60% in feces. Plasma concentrations of total radioactivity (simvastatin plus ^{14}C-metabolites) peaked at 4 hours and declined rapidly to about 10% of peak by 12 hours post dose. Since simvastatin undergoes extensive first-pass extraction in the liver, the availability of the drug to the general circulation is low (<5%).

Both simvastatin and its β-hydroxyacid metabolite are highly bound (approximately 95%) to human plasma proteins. Simvastatin crosses the blood-brain barrier on administration of radiolabeled simvastatin in rats.

The major active metabolites of simvastatin present in human plasma are the β-hydroxyacid of simvastatin and its 6'-hydroxy, 6'-hydroxymethyl, and 6'-exomethylene derivatives. Peak plasma concentrations of both active and total inhibitors were attained within 1.3 to 2.4 hours post dose. There was no substantial deviation from linearity of AUC of inhibitors in the general circulation with an increase in dose to as high as 120 mg. Concomitant food intake does not affect the absorption of simvastatin.

The mean plasma level of HMG-CoA reductase inhibitory activity is approximately 45% higher in elderly patients (age 70 - 78 yrs) compared to younger patients (age 18-30 yrs). There are no overall differences in safety between elderly and younger patients.

Kinetic studies reported with another statin, having a similar principal route of elimination, have suggested that for a given dose level higher systemic exposure may be achieved in patients with severe renal insufficiency (as measured by creatinine clearance).

Although the mechanism is not fully understood, cyclosporine has been shown to increase the AUC of statins. The increase in AUC for simvastatin acid is presumably due, in part, to inhibition of CYP3A4.

The risk of myopathy is increased by high levels of HMG-CoA reductase inhibitory activity in plasma. Inhibitors of CYP3A4 can raise the plasma levels of HMG-CoA reductase inhibitory activity and increase the risk of myopathy (see WARNINGS AND PRECAUTIONS).

Simvastatin (at dose 80-mg) has no effect on the metabolism of the probe cytochrome P450 isoform 3A4 (CYP3A4) substrates midazolam and erythromycin in healthy subjects.

Coadministration of simvastatin (40 mg QD for 10 days) resulted in an increase in the maximum mean levels of cardioactive digoxin (given as a single 0.4 mg dose on day 10) by approximately 0.3 ng/mL.

INDICATIONS / USAGE

Reductions in risk of CHD mortality and cardiovascular events

In patients at high risk of coronary events because of existing coronary heart disease, diabetes, peripheral vessel disease, history of stroke or other cerebrovascular disease, simvastatin is indicated to:

- Reduce the risk of total mortality by reducing CHD deaths.
- Reduce the risk of non-fatal myocardial infarction and stroke.
- Reduce the need for coronary and non-coronary revascularization procedures

Hyperlipidemia

Simvastatin is indicated as an adjunct to diet, to reduce elevated total cholesterol (total-C), low-density lipoprotein cholesterol (LDL-C), apolipoprotein B (Apo B), and triglycerides (TG), and to increase high-density lipoprotein cholesterol (HDL-C) in patients with primary hypercholesterolemia, heterozygous familial hypercholesterolemia or combined (mixed) dyslipidemia when response to diet and other non-pharmacological measures is inadequate. Simvastatin therefore, lowers the LDL-C/HDL-C and total-C/HDL-C ratios.

Pediatric patient with heterozygous familial hypercholesterolemia

Simvastatin is indicated as an adjunct to diet to reduce total-C, LDL-C, TG, and Apo B level in adolescent boys and girls who are at least one year post-menarche, 10-17 years of age with heterozygous familial hypercholesterolemia (HeFH).

General recommendation

Prior to initiating therapy with simvastatin, secondary causes of hypercholesterolemia (e.g. Poorly controlled diabetes mellitus, hypothyroidism, nephrotic syndrome, dysproteinemias, obstructive liver disease, other drug therapy, alcoholism) should be identified and treated.

DOSE AND METHOD OF ADMINISTRATION

Prior to initiating therapy with simvastatin, secondary causes of hypercholesterolemia (e.g. poorly controlled diabetes mellitus, hypothyroidism, nephrotic syndrome, dysproteinemias, obstructive liver disease, other drug therapy, alcoholism) should be identified and treated.

The dosage range for simvastatin is 5-80 mg/day, given as a single dose in the evening. Adjustments of dosage, if required, should be made at intervals of not less than 4 weeks, to a maximum of 80 mg/day given as a single dose in the evening. The 80-mg dose of simvastatin is only recommended in patients at high risk for cardiovascular complications who have not achieved treatment goals on lower doses and when the benefits are expected to outweigh the potential risks. The recommended usual starting dose is 20-40 mg once a day in the evening.

Patients at high risk of coronary heart disease (CHD) or with existing CHD

The usual starting dose of simvastatin is 40 mg/day given as a single dose in the evening in patients at high risk of CHD (with or without hyperlipidemia), ie., patients with diabetes, history of stroke or other cerebrovascular disease, peripheral vessel disease, or with existing CHD. Drug therapy can be initiated simultaneously with a standard cholesterol-lowering diet and exercise.

Patients with hyperlipidemia (who are not in the risk categories above)

The patient should be placed on a standard cholesterol-lowering diet before receiving simvastatin and should continue on this diet during treatment with simvastatin. The usual starting dose is 20 mg/day given as a single dose a day in the evening. Patients who require a large reduction in LDL-C (more than 45%) may be started at 40mg/day given as single dose in the evening. Patients with mild to moderate hypercholesterolemia can be treated with a starting dose of 10mg of simvastatin. Adjustments of dosage, if required, should be made as specified above.

Homozygous familial hypercholesterolemia

The recommended dosage for patients with homozygous familial hypercholesterolemia is simvastatin 40 mg/day in the evening. The 80-mg dose is only recommended when the benefits are expected to outweigh the potential risks. Simvastatin should be used as an adjunct to other lipid-lowering treatments (e.g., LDL apheresis) in these patients or if such treatments are unavailable.

Concomitant therapy
Simvastatin is effective alone or in combination with bile acid sequestrants.

- In patients taking simvastatin concomitantly with fibrates (other than gemfibrozil or fenofibrate), the dose of simvastatin should not exceed 10 mg/day.
- In patients taking amiodarone, verapamil or diltiazem concomitantly with simvastatin, the dose of simvastatin should not exceed 20mg/day.
- In patients taking amlodipine or lipid-lowering dose of niacin (≥1 g/day) concomitantly with simvastatin, the dose of simvastatin should not exceed 40 mg/day.

Dosage in renal insufficiency
Because simvastatin does not undergo significant renal excretion, modification of dosage should not be necessary in patients with moderate renal insufficiency. In patients with severe renal insufficiency (creatinine clearance <30 mL/min), dosages above 10 mg/day should be carefully considered and, if deemed necessary, implemented cautiously.

Dosage in pediatric patients (10-17 years of age) with heterozygous familial hypercholesterolemia
The recommended usual starting dose is 10 mg once a day in the evening. The recommended dosing range is 10-40mg/day; the maximum recommended dose is 40 mg/day. Doses should be individualized according to the recommended goal of therapy.

CONTRAINDICATIONS

SIMVOR (simvastatin) is contraindicated in the following conditions:

- Hypersensitivity to any component of this medication

- Active liver disease, which may include unexplained persistent elevations in hepatic transaminase levels (see WARNINGS AND PRECAUTIONS).

- Pregnancy and lactation (see USE IN SPECIAL POPULATIONS*)*.

- Concomitant administration of strong CYP3A4 inhibitors (e.g., itraconazole, ketoconazole, posaconazole, voriconazole, HIV protease inhibitors, boceprevir, telaprevir, erythromycin, clarithromycin, telithromycin, cobicistat-containing products and nefazodone) [see WARNINGS AND PRECAUTIONS].

- Concomitant administration of gemfibrozil, cyclosporine, or danazol (see WARNINGS AND PRECAUTIONS)

WARNINGS AND PRECAUTIONS

Myopathy/Rhabdomyolysis

Simvastatin, like other inhibitors of HMG-CoA reductase, occasionally causes myopathy manifested as muscle pain, tenderness or weakness with creatine kinase (CK) above ten times the upper limit of normal (ULN). Myopathy sometimes takes the form of rhabdomyolysis with or without acute renal failure secondary to myoglobinuria, and rare fatalities have occurred. The risk of myopathy is increased by high levels of HMG-CoA reductase inhibitory activity in plasma. Predisposing factors for myopathy include advanced age (≥65 years), female gender, uncontrolled hypothyroidism, personal or familial history of hereditary muscular disorders, previous history of muscular toxicity with a statin or fibrate, alcohol abuse and renal impairment.

As with other HMG-CoA reductase inhibitors, the risk of myopathy/rhabdomyolysis is dose related. Approximately 0.03%, 0.08% and 0.61% incidence of myopathy is reported at 20, 40 and 80 mg/day, respectively in patients treated with simvastatin.

The risk of myopathy, including rhabdomyolysis, is greater in patients on simvastatin 80 mg compared with other statin therapies with similar or greater LDL-C-lowering efficacy and compared with lower doses of simvastatin. Therefore, the 80mg dose is only recommended in patients at high risk for cardiovascular complications who have not achieved treatment goals on lower doses and when the benefits are expected to outweigh the potential risks. Patients should be advised of the increased risk of myopathy, including rhabdomyolysis, and to report promptly any unexplained muscle pain, tenderness or weakness. If symptoms occur, treatment should be discontinued immediately.

All patients starting therapy with simvastatin, or whose dose of simvastatin is being increased, should be advised of the risk of myopathy, including rhabdomyolysis and told to report promptly any unexplained muscle pain, tenderness or weakness. Simvastatin therapy should be discontinued immediately if myopathy is diagnosed or suspected. In most cases, muscle symptoms and CK increases resolved when treatment was promptly discontinued. Periodic CK determinations may be considered in patients starting therapy with simvastatin or whose dose is being increased, but there is no assurance that such monitoring will prevent myopathy.

Many of the patients who have developed rhabdomyolysis on therapy with simvastatin have had complicated medical histories, including renal insufficiency usually as a consequence of long-standing diabetes mellitus. Such patients merit closer monitoring. Therapy with simvastatin should be discontinued if markedly elevated CPK levels occur or myopathy is diagnosed or suspected. Simvastatin therapy should also be temporarily withheld in any patient experiencing an acute or serious condition predisposing to the development of renal failure secondary to rhabdomyolysis, e.g., sepsis; hypotension; major surgery; trauma; severe metabolic, endocrine, or electrolyte disorders; or uncontrolled epilepsy.

The risk of myopathy and rhabdomyolysis is increased by high levels of statin activity in plasma. Simvastatin is metabolized by the cytochrome P450 isoform 3A4. Certain drugs which inhibit this metabolic pathway can raise the plasma levels of simvastatin and may increase the risk of myopathy. These include itraconazole, ketoconazole, posaconazole, voriconazole, the macrolide antibiotics erythromycin and clarithromycin, and the ketolide antibiotic telithromycin, HIV protease inhibitors, boceprevir, telaprevir, the antidepressant nefazodone, cobicistat-containing products or grapefruit juice. Combination of these drugs with simvastatin is contraindicated. If short-term treatment with strong CYP3A4 inhibitors is unavoidable, therapy with simvastatin must be suspended during the course of treatment. (see DRUG INTERACTIONS).

The combined use of simvastatin with gemfibrozil, cyclosporine, or danazol is contraindicated (see CONTRAINDICATIONS and DRUG INTERACTIONS).

Caution should be used when prescribing other fibrates with simvastatin, as these agents can cause myopathy when given alone and the risk is increased when they are co-administered (see DRUG INTERACTIONS).

Cases of myopathy, including rhabdomyolysis, have been reported with simvastatin coadministered with colchicine, and caution should be exercised when prescribing simvastatin with colchicine (see DRUG INTERACTIONS).

There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statin. IMNM is clinically characterized by:

- persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment;
- muscle biopsy showing necrotizing myopathy without significant inflammation
- improvement with immunosuppressive agents.

The benefits of the combined use of simvastatin with the following drugs should be carefully weighed against the potential risks of combinations: other lipid-lowering drugs (other fibrates, ≥ 1 g/day of niacin), amiodarone, dronedarone, verapamil, diltiazem, amlodipine, or ranolazine (see DRUG INTERACTIONS).

Prescribing recommendations for interacting agents are summarized in Table below:

Table: Drug Interactions Associated with Increased Risk of Myopathy/Rhabdomyolysis	
Interacting Agents	Prescribing Recommendations
Itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors, nefazodone gemfibrozil, cyclosporine, danazol, boceprevir, telaprevir, cobicistat-containing products	Contraindicated with simvastatin
Amiodarone, verapamil, diltiazem	Do not exceed 20 mg simvastatin daily
Dronedarone	Do not exceed 10 mg simvastatin daily
Ranolazine	Do not exceed 20 mg simvastatin daily
Amlodipine	Do not exceed 40 mg simvastatin

Table: Drug Interactions Associated with Increased Risk of Myopathy/Rhabdomyolysis	
Interacting Agents	Prescribing Recommendations
	daily
Other Fibrates (except fenofibrate)	Do not exceed 10 mg simvastatin daily
Fusidic Acid	Is not recommended with simvastatin
Grapefruit juice	Avoid grapefruit juice

A higher than expected incidence of myopathy has been reported in Chinese patients taking simvastatin 40 mg and nicotinic acid/laropiprant 2000 mg/40 mg. Therefore, caution should be used when treating Chinese patients with simvastatin (particularly doses of 40 mg or higher) co-administered with lipid-modifying doses (≥1 g/day) of niacin (nicotinic acid) or products containing niacin. Because the risk of myopathy with statins is dose-related, the use of simvastatin 80 mg with lipid-modifying doses (≥1 g/day) of niacin (nicotinic acid) or products containing niacin is not recommended in Chinese patients. It is unknown whether there is an increased risk of myopathy in other Asian patients treated with simvastatin co-administered with lipid-modifying doses (≥ 1 g/day) of niacin (nicotinic acid) or products containing niacin.

If the combination proves necessary, patients on fusidic acid and simvastatin should be closely monitored. Temporary suspension of simvastatin treatment may be considered.

Liver Dysfunction

Persistent increases (to more than three times the ULN) in serum transaminases have been reported in approximately 1% of patients who received simvastatin in clinical studies. When drug treatment was interrupted or discontinued in these patients, the transaminase levels usually fell slowly to pretreatment levels. The increases were not associated with jaundice or other clinical signs or symptoms. There was no evidence of hypersensitivity.

There was no significant difference in transaminase elevation (to >3X ULN) between the simvastatin and placebo groups (0.7% vs. 0.6%) in patients with coronary heart disease in a reported double-blind randomized clinical trial. Elevated transaminases resulted in the discontinuation of 8 patients from therapy in the simvastatin group and 5 in the placebo group. Of the simvastatin treated patients with baseline normal liver function tests (LFTs), 0.4% patients developed consecutive >3X ULN of liver function test and/or were discontinued due to transaminase elevations during the 5.4 years. All of the patients in this study received a starting dose of 20 mg of simvastatin; 37% were titrated to 40 mg.

It has been reported that 12-month incidence of persistent hepatic transaminase elevation without regard to drug relationship was 0.9% and 2.1% at 40- and 80-mg dose, respectively. No patients developed persistent liver function abnormalities following the initial 6 months of treatment at a given dose.

It is recommended that liver function tests be performed before the initiation of treatment, and thereafter when clinically indicated. Patients who develop increased transaminase levels should be monitored with a second liver function evaluation to confirm the finding and be followed thereafter with frequent liver function tests until the abnormality (ies) return to normal. Should an increase in

AST or ALT of 3X ULN or greater persist, withdrawal of therapy with simvastatin is recommended. Note that ALT may emanate from muscle, therefore ALT rising with CK may indicate myopathy.

The drug should be used with caution in patients who consume substantial quantities of alcohol and/or have a past history of liver disease. Active liver diseases or unexplained transaminase elevations are contraindications to the use of simvastatin.

As with other lipid-lowering agents, moderate (less than 3X ULN) elevations of serum transaminases have been reported following therapy with simvastatin. These changes appeared soon after initiation of therapy with simvastatin, were often transient, were not accompanied by any symptoms and did not require interruption of treatment.

Diabetes mellitus

Some evidence suggests that statins as a class raise blood glucose and in some patients, at high risk of future diabetes, may produce a level of hyperglycaemia where formal diabetes care is appropriate. This risk, however, is outweighed by the reduction in vascular risk with statins and therefore should not be a reason for stopping statin treatment. Patients at risk (fasting glucose 5.6 to 6.9 mmol/L, BMI > 30 kg/m^2, raised triglycerides, hypertension) should be monitored both clinically and biochemically according to national guidelines

Interstitial lung disease

Cases of interstitial lung disease have been reported with some statins, including simvastatin especially with long term therapy. Presenting features can include dyspnoea, non-productive cough and deterioration in general health (fatigue, weight loss and fever). If it is suspected a patient has developed interstitial lung disease, statin therapy should be discontinued.

Ophthalmic Evaluations

In the absence of any drug therapy, an increase in the prevalence of lens opacities with time is expected as a result of aging. Current long-term data from clinical studies do not indicate an adverse effect of simvastatin on the human lens.

Effects on ability to drive and use machines

Simvaststin has no or negligible influence on the ability to drive and use machines. However, when driving vehicles or operating machines, it should be taken into account that dizziness has been reported rarely in post-marketing experiences.

SIMVOR tablets contain lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicine.

USE IN SPECIAL POPULATIONS

- Pregnancy

Pregnancy Category X (see CONTRAINDICATONS).

Simvastatin is contraindicated in women who are or may become pregnant. Lipid lowering drugs offer no benefit during pregnancy, because cholesterol and cholesterol derivatives are needed for normal fetal development. Atherosclerosis is a chronic process, and discontinuation of lipid-lowering drugs during pregnancy should have little impact on long-term outcomes of primary hypercholesterolemia therapy. There are no adequate and well-controlled studies of use with Simvastatin during pregnancy; however, there are rare reports of congenital anomalies in infants exposed to statins *in utero*. Simvastatin was not teratogenic in rats or rabbits at doses (25, 10 mg/kg/day, respectively) that resulted in 3 times the human exposure based on mg/m^2 surface area. However, in studies with another structurally-related statin, skeletal malformations were observed in rats and mice. Serum cholesterol and triglycerides increase during normal pregnancy, and cholesterol or cholesterol derivatives are essential for fetal development. Because statins decrease cholesterol synthesis and possibly the synthesis of other biologically active substances derived from cholesterol, simvastatin may cause fetal harm when administered to a pregnant woman. If simvastatin is used during pregnancy or if the patient becomes pregnant while taking this drug, the patient should be apprised of the potential hazard to the fetus.

There are rare reports of congenital anomalies following intrauterine exposure to statins. In a review of approximately 100 prospectively followed pregnancies in women exposed to simvastatin or another structurally related statin, the incidences of congenital anomalies, spontaneous abortions, and fetal deaths/stillbirths did not exceed those expected in the general population. However, the study was only able to exclude a 3- to 4-fold increased risk of congenital anomalies over the background rate. In 89% of these cases, drug treatment was initiated prior to pregnancy and was discontinued during the first trimester when pregnancy was identified.

Women of childbearing potential, who require treatment with simvastatin for a lipid disorder, should be advised to use effective contraception. For women trying to conceive, discontinuation of simvastatin should be considered. If pregnancy occurs, simvastatin should be immediately discontinued.

- Lactation

It is not known whether simvastatin is excreted in human milk. Because a small amount of another drug in this class is excreted in human milk and because of the potential for serious adverse reactions in nursing infants, women taking simvastatin should not nurse their infants. A decision should be made whether to discontinue nursing or discontinue drug, taking into account the importance of the drug to the mother (see CONTRAINDICATONS).

- Pediatrics

Safety and effectiveness of simvastatin in patients 10-17 years of age with heterozygous familial hypercholesterolaemia have been reported in adolescent boys (Tanner Stage II and above) and in girls who were at least one year post-menarche. Patients treated with simvastatin had an adverse experience profile generally similar to that of patients receiving placebo. Doses greater than 40 mg have not been studied in this population. It has been reported that there was no detectable effect on growth or sexual maturation in the adolescent boys or girls, or any effect on menstrual cycle length in girls. Adolescent females should be counselled on appropriate contraceptive methods while on simvastatin therapy. In patients aged < 18 years, efficacy and safety have not been studied for treatment periods > 48 weeks' duration and long-term effects on physical, intellectual, and sexual maturation are unknown. Simvastatin has not been studied in patients younger than 10 years of age, nor in pre-pubertal children and pre- menarchal girls.

- Geriatrics

No overall differences in safety or effectiveness have been reported between the elderly and younger patients, but greater sensitivity of some older individuals cannot be ruled out. Since advanced age (≥65 years) is a predisposing factor for myopathy, simvastatin should be prescribed with caution in the elderly (see Pharmacokinetics).

- Renal impairment

Caution should be exercised when simvastatin is administered to patients with severe renal impairment.

- Hepatic impairment

Simvastatin is contraindicated in patients with active liver disease which may include unexplained persistent elevations in hepatic transaminase levels.

DRUG INTERACTIONS

Contraindicated Drugs

Potent inhibitors of CYP3A4: Concomitant use with medicines labeled as having a potent inhibitory effect on CYP3A4 at therapeutic doses (e.g.: itraconazole, ketoconazole, posaconazole, voriconazole, erythromycin, clarithromycin, telithromycin, HIV protease inhibitors, boceprevir, telaprevir or nefazodone) is contraindicated. If treatment with potent CYP3A4 inhibitors is unavoidable, therapy with simvastatin should be suspended during the course of treatment.

Gemfibrozil, cyclosporine or danazol: Concomitant use of these drugs with simvastatin is contraindicated.

Moderate inhibitors of CYP3A4

Patients taking other medicines labeled as having a moderate inhibitory effect on CYP3A4 concomitantly with simvastatin, particularly higher simvastatin doses, may have an increased risk of myopathy. When coadministering simvastatin with a moderate inhibitor of CYP3A4, a dose adjustment of simvastatin may be necessary.

Lipid-Lowering Drugs That Can Cause Myopathy When Given Alone

Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Other fibrates:

The dose of simvastatin should not exceed 10 mg daily in patients receiving concomitant medication with fibrates other than gemfibrozil (see CONTRAINDICATIONS) or fenofibrate. When simvastatin and fenofibrate are given concomitantly, there is no evidence that the risk of myopathy exceeds the sum of the individual risks of each agent. Caution should be used when prescribing fenofibrate with simvastatin, as either agent can cause myopathy when given alone. Addition of fibrates to simvastatin typically provides little additional reduction in LDL-C, but further reductions of TG and further increases in HDL-C may be obtained. Combinations of fibrates with simvastatin have been used without myopathy in small short-term clinical studies with careful monitoring. (see WARNINGS AND PRECAUTIONS).

Amiodarone:

In a clinical trial, myopathy was reported in 6% of patients receiving simvastatin 80 mg and amiodarone. The dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with amiodarone.

Calcium channel blockers:

- Verapamil or diltiazem: In a clinical trial, patients on diltiazem treated concomitantly with simvastatin 80 mg had an increased risk of myopathy. The dose of simvastatin should not exceed 20 mg daily in patients receiving concomitant medication with verapamil or diltiazem.

- Amlodipine: In a clinical trial, patients on amlodipine treated concomitantly with simvastatin 80 mg had a slightly increased risk of myopathy. The dose of simvastatin should not exceed 40 mg daily in patients receiving concomitant medication with amlodipine.

Niacin (≥1g/day):
The dose of simvastatin should not exceed 40mg daily in patients receiving concomitant medication with niacin (nicotinic acid) ≥ 1g/day. Cases of myopathy/rhabdomyolysis have been observed with simvastatin co-administered with lipid-modifying doses (≥ 1 g/day) of niacin.

Dronedarone or Ranolazine
The risk of myopathy, including rhabdomyolysis, is increased by concomitant administration of dronedarone or ranolazine.

Digoxin
Concomitant administration of digoxin with simvastatin resulted in a slight elevation in digoxin concentrations in plasma. Patients taking digoxin should be monitored appropriately when simvastatin is initiated.

Coumarin Anticoagulants
Simvastatin 20-40 mg/day is reported to modestly potentiate the effect of coumarin anticoagulants in normal volunteers and in hypercholesterolemic patients. Prothrombin time, reported as International Normalized Ratio (INR), increased from a baseline of 1.7 to 1.8 and from 2.6 to 3.4 in the volunteers and hypercholesterolemic patients, respectively. With other statins, clinically evident bleeding and/or increased prothrombin time has been reported in a few patients taking coumarin anticoagulants concomitantly. In such patients, prothrombin time should be determined before starting simvastatin and frequently enough during early therapy to ensure that no significant alteration of prothrombin time occurs. Once a stable prothrombin time has been documented, prothrombin times can be monitored at the intervals usually recommended for patients on coumarin anticoagulants. If the dose of simvastatin is changed or discontinued, the same procedure should be repeated. Simvastatin therapy has not been associated with bleeding or with changes in prothrombin time in patients not taking anticoagulants.

Colchicine
Cases of myopathy, including rhabdomyolysis, have been reported with simvastatin coadministered with colchicine, and caution should be exercised when prescribing simvastatin with colchicine.

Fluconazole

Rare cases of rhabdomyolysis associated with concomitant administration of simvastatin and fluconazole have been reported.

Fusidic acid
Patients on fusidic acid treated concomitantly with simvastatin may have an increased risk of myopathy/rhabdomyolysis).

Co-administration with fusidic acid is not recommended. In patients where the use of systemic fusidic acid is considered essential, simvastatin should be discontinued throughout the duration of fusidic acid treatment. In exceptional circumstances, where prolonged systemic fusidic acid is needed, e.g. for the treatment of severe infections, the need for co-administration of simvastatin and fusidic acid should only be considered on a case-by-case basis under close medical supervision (see WARNINGS AND PRECAUTIONS).

Grapefruit juice
Grape fruit juice contains one or more components that inhibit CYP3A4 and can increase the plasma levels of drug metabolized by CYP3A4. The effect of typical consumption (one 250-ml glass daily) is minimal (13 % increase in active plasma HMG-CoA reductase inhibitory activity as measured by the area under concentration-time curve) and of no clinical relevance. However, very large quantities (over 1 liter daily) significantly increase plasma level of HMG-CoA reductase inhibitory activity during simvastatin therapy and should be avoided.

Rifampicin
Because rifampicin is a potent CYP3A4 inducer, patients undertaking long-term rifampicin therapy (e.g. treatment of tuberculosis) may experience loss of efficacy of simvastatin. It has been reported that the area under the plasma concentration curve (AUC) for simvastatin acid was decreased by 93% with concomitant administration of rifampicin in normal volunteers.

HIV protease inhibitors (see CONTRAINDICATIONS and WARNINGS AND PRECAUTIONS).

Antidepressant nefazodone (see CONTRAINDICATIONS and WARNINGS AND PRECAUTIONS).

SIDE EFFECTS/UNDESIRABLE EFFECTS

The frequencies of adverse events are ranked according to the following: Very common (> 1/10), Common (≥ 1/100, < 1/10), Uncommon (≥ 1/1000, < 1/100), Rare (≥ 1/10,000, < 1/1000), Very Rare (< 1/10,000), not known (cannot be estimated from the reported data).

<u>Blood and lymphatic system disorders:</u>
Rare: anaemia
<u>Immune system disorders:</u>
Very rare: anaphylaxis
<u>Psychiatric disorders:</u>
Very rare: insomnia
Not known: depression

Nervous system disorders:

Rare: headache, paresthesia, dizziness, peripheral neuropathy

Very rare: memory impairment

Respiratory, thoracic and mediastinal disorders:

Not known: interstitial lung disease

Gastrointestinal disorders:

Rare: constipation, abdominal pain, flatulence, dyspepsia, diarrhoea, nausea, vomiting, pancreatitis

Hepatobiliary disorders:

Rare: hepatitis/jaundice

Very rare: fatal and non-fatal hepatic failure

Skin and subcutaneous tissue disorders:

Rare: rash, pruritus, alopecia

Musculoskeletal and connective tissue disorders:

Rare: myopathy* (including myositis), rhabdomyolysis with or without acute renal failure, myalgia, muscle cramps

* Myopathy has been reported to occur commonly in patients treated with simvastatin 80 mg/day compared to patients treated with 20 mg/day.

Not known: tendinopathy, sometimes complicated by rupture; immune-mediated necrotizing myopathy (IMNM)**

** There have been very rare reports of immune-mediated necrotizing myopathy (IMNM), an autoimmune myopathy, during or after treatment with some statins. IMNM is clinically characterized by: persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment; muscle biopsy showing necrotizing myopathy without significant inflammation; improvement with immunosuppressive agents.

Reproductive system and breast disorders:

Not known: erectile dysfunction

General disorders and administration site conditions:

Rare: asthenia

An apparent hypersensitivity syndrome has been reported rarely which has included some of the following features: angioedema, lupus-like syndrome, polymyalgia rheumatica, dermatomyositis, vasculitis, thrombocytopenia, eosinophilia, ESR increased, arthritis and arthralgia, urticaria, photosensitivity, fever, flushing, dyspnoea and malaise.

Investigations:

Rare: increases in serum transaminases (alanine aminotransferase, aspartate aminotransferase, γ-glutamyl transpeptidase), elevated alkaline phosphatase; increase in serum CK levels.

Increases in HbA1c and fasting serum glucose levels have been reported with statins, including simvastatin.

There have been rare reports of cognitive impairment (e.g., memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use, including simvastatin. The reports are generally nonserious, and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median of 3 weeks).

The following additional adverse events have been reported with some statins:
• Sleep disturbances, including nightmares
• Sexual dysfunction
• Diabetes mellitus: Frequency will depend on the presence or absence of risk factors (fasting blood glucose ≥ 5.6 mmol/L, BMI > 30 kg/m^2, raised triglycerides, history of hypertension).

<u>Paediatric population</u>
In a study involving children and adolescents (boys Tanner Stage II and above and girls who were at least one year post-menarche) 10-17 years of age with heterozygous familial hypercholesterolaemia,

The safety and tolerability profile of the group treated with simvastatin was generally similar to that of the group treated with placebo. The long-term effects on physical, intellectual, and sexual maturation are unknown. No sufficient data are currently reported after one year of treatment

The safety and tolerability profile of simvastatin in children and adolescents (boys Tanner Stage II and above and girls who were at least one year post-menarche) 10-17 years of age with heterozygous familial hypercholesterolaemia has been reported to be generally similar to that of placebo. The long-term effects on physical, intellectual, and sexual maturation are not reported. No sufficient data are currently reported after one year of treatment.

OVERDOSE

A few cases of overdosage with simvastatin have been reported; the maximum dose taken was 3.6 g. All patients recovered without sequelae. Supportive measures should be taken in the event of an overdose. The dialyzability of simvastatin and its metabolites in man is not known at present.

STORAGE
Store below 30^{o}C, protected from moisture

PACKING
SIMVOR TABLETS 10 mg:
Blisters of 3 x 10's, 5 x 10's, 10x10's

SIMVOR TABLETS 20 mg:
Blisters of of 3 x 10's, 5 x 10's, 10x10's

SIMVOR TABLETS 40 mg:
Blisters of 10 x 10's and 3 x 10's

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